5 A cell P, a fixed resistor R and a uniform resistance wire AB are connected in a circuit as shown in Fig. 5.1.

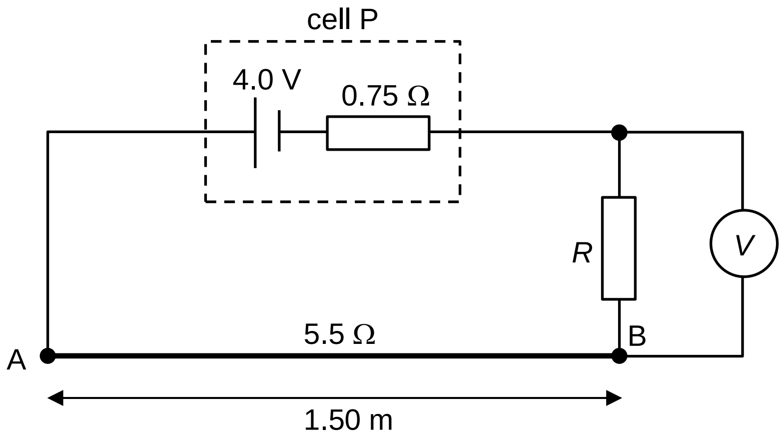

Fig. 5.1

Cell P has e.m.f. 4.0 V and internal resistance 0.75 Ω. Wire AB has length 1.50 m and resistance 5.5 Ω. The voltmeter reads 1.3 V.

(a) Show that the potential difference across AB is 2.4 V. [2]

p.d. across R = 1.30 V

Using potential divider rule,

$$\frac{R}{0.75 + R + 5.5} \times 4.0 = 1.3$$

$$R = 3.0\ \Omega$$

Hence, p.d. across AB $= \dfrac{5.5}{0.75 + 3.0 + 5.5} \times 4.0$

$= 2.4$ V

(b) A cell Q and a sensitive ammeter are connected to the circuit in Fig. 5.1, as shown in Fig. 5.2.

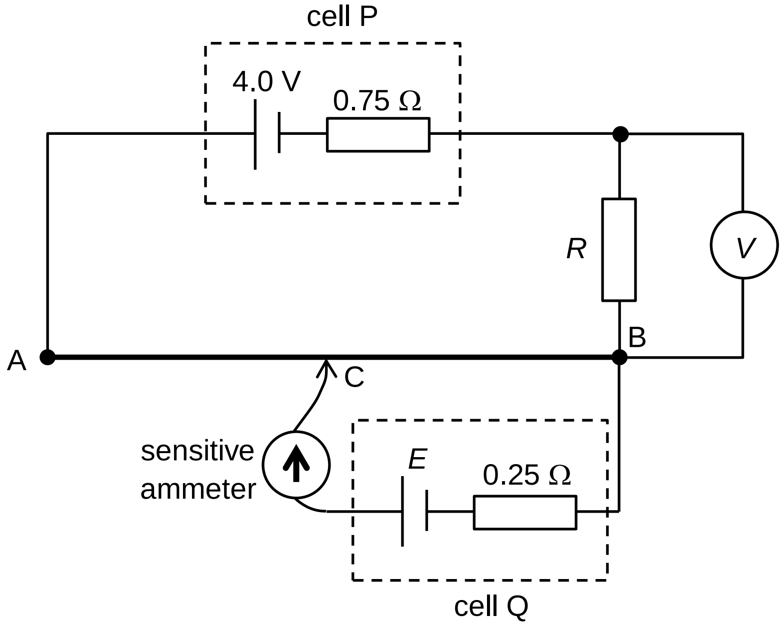

Fig. 5.2

Cell Q has e.m.f. *E* and internal resistance 0.25 Ω. The ammeter reads zero when the length of AC is 0.56 m.

(i) Determine *E*.

$$E = \frac{1.50 - 0.56}{1.50} \times 2.4$$
$$= 1.5 \text{ V}$$

E = ... V [2]

(ii) There is a reading on the ammeter when the connection C is shifted closer to A. State and explain the direction of the current across cell Q.

The potential at C is higher than that at B. As C is shifted closer to A, the potential at C increases, thus increasing the potential difference between BC. [1]

Since the potential between BC will become larger than the terminal p.d. of cell Q, current will now flow through cell Q from C to B. [1]

....[2]

(d) The resistance wire AB is detached from the circuit and coiled in a circular manner as shown in Fig. 5.3. Metallic fasteners with negligible resistance are used to secure the wire at X and Y, where XY is the diameter of the coil.

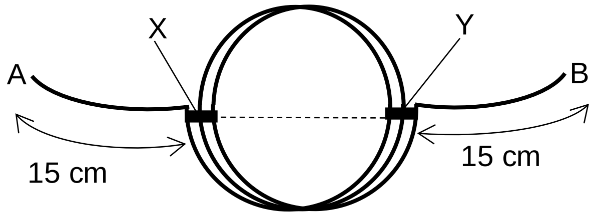

Fig. 5.3

(i) Determine the resistance of wire AB when coiled in this manner.

$$\frac{R}{L} = 5.5/150 = 0.0367\ \Omega\ \text{cm}^{-1}\ [1]$$

$$R = 2 \times 15 \times 0.0367 + \frac{1}{5}\left(\frac{150 - 2 \times 15}{5} \times 0.0367\right)\ [1]$$

$$= 1.28\ \Omega$$

$$= 1.3\ \Omega\ [1]$$

resistance = ... Ω [3]

(ii) An e.m.f. source is again connected across AB. State and explain whether the drift velocity of the electrons is greater in AX or in XY.

Since $I = n\,A\,v\,q$, drift velocity v is greater in AX than in XY

Method 1: as the current I is greater in AX than in either parallel section of XY.

Method 2: as the combined cross sectional area of XY is greater than AX. [2]

[Total: 11]

6 W is a long straight wire of length 1.2 m. A beam of electrons is projected with a speed of